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Social media sentiment analysis through parallel dilated convolutional neural network for smart city applications

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ABSTRACT

Deep Learning is considered to leverage smart cities through social media sentiment analysis. The digital content in social media can be used for many smart city applications (SCAs)[1]. Classical convolutional neural networks (CNNs) are challenging to parallelize and insufficient to capture long term contextual semantic features for sentiment analysis. In this perspective, this paper initially proposes a domain-specific distributed word representation (DS-DWR)[2] with a considerably small corpus size induced from textual resources in social media. In DS-DWR, different Distributed Word Representations are concatenated to builds rich representations over the input sequence, which is worthwhile for infrequent and unseen terms. Second, a dilated convolutional neural network (D-CNN)[3], which is composed of three parallel dilated convolutional neural network (PD-CNN)[4] layers and a global average pooling (GAP)[5] layer. Our considered parallel dilated convolution reduces dimension and incorporates an extension in the size of receptive fields without the loss of local information. Further, the long-term contextual semantic information is achieved by the use of different dilation rates. Experiments demonstrate that our architecture accomplishes comparable results with multiple hyperparameters tuning for better parallelism which leads to the minimized computational cost.

1. Introduction

The development of smart city applications (SCAs) in the last decade has shown flexible design and valuable services. Smart Cities are a combination of technical and social advancements, e.g., Internet access on Mobile, Social Data Analytics, and Internet of Things (IoT) [1]. These advancements can facilitate to manage resources productively primarily through social networks. Our study endorses a novel way for the smart cities towards the building of intelligent and adaptable SCAs. The social media has become a potential information source for social information mining to dominate people's opinions. In our perspective, users can be viewed as social sensors, and opinions are response signals as it is real-time characteristics of social data. These opinions are extensive on social media consist of short sentences and contain useful information in numerous contexts. Therefore, social media data as social sensors can be employed to discover valuable information. A developing pattern in the area of social data mining is to concentrate on social textual data having opinions instead of concentrating on extensive numerical analysis. Social media sentiment analysis is a growing technique to comprehend the opinions of individuals through social networks. While SCAs can utilize the benefits of social media sentiment analysis as the opinions can be related to any event or a source of estimating preferences, dislikes, and patterns.

SCAs can act as "activators" by focusing on improvements in technological, political, and organizational aspects through social media. Also, it can enhance people's attitudes and empower them to build up a feasible environment in Smart cities [2]. Social networks are the primary source of opinions and events; however, the main issue is substantial social content that requires smart approaches to filter out noisy data. The prerequisite of noise filtering of social media data involves automatic classification techniques for valuable facts. It also suffers a few difficulties; for example, sentences are short and contain abbreviations as well as spelling mistakes. The semantic analysis can be used to manage abundant social media data to induce semantic for improvements in terms of integration and reusability [3]. Similarly,

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[1] Smart city applications
[2] Domain Specific distributed word representation
[3] Dilated Convolutional Neural Network
[4] Parallel dilated convolutional neural network
[5] Global Average Pooling

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social media mining techniques can utilize to collect and process social data posts on Facebook, Twitter, and Instagram for SCAs [4]. We choose Twitter, which has a considerable number of posts and consists of less than 140 characters [5]. This social media data can be used for social media sentiment analysis and considered appropriate to examine people's opinions of smart city services.

Currently, Machine Learning (ML) methodologies utilized IoT and Big Data for the enhancements of smart city services [6]. Among many ML methods on classifying text, Naıve Bayes (NB) is utilized for spam detection [7], topic detection [8], sentiment analysis [9], recommendation systems [10]. In a social context, popular techniques support vector machine (SVM) has been utilized to characterize tweets to get the trends [11]. However, there are some issues related to the extraction of the features with variable-size sequences composition. Whereas a subfield, deep Learning (DL), favorably provides intelligent services. DL employs artificial neural networks (ANNs) architectures in order to learn high-level features for social data with multiple layers. These features are efficient enough to replace many ML methodologies, especially when addressing social media data for predictions. Further, these methodologies are progressively applied to supervised and unsupervised classification problems [12–14].

Deep learning (DL) persuades researchers by using ANNs that empowers efficient learning of features without requiring complex feature-engineering [15,16]. DL methodologies work by executing the feature extraction and classification task through initiating the representation of a sequence of words via multiplying with associated weight matrix as a one-hot vector [17]. This sequence of each word is formulated by continuous vector space inputted to a neural network to process the succession of words with the assistance of various layers for predication. This processing has an impact on the training set towards increasing the classification accuracy, as explained in [18]. Convolutional neural network (CNN) has accomplished excellent outcomes of sentence-level classification [19,20]. It can learn from different distributed word representations such as Word2Vec [21], FastText [22] and GloVe [23] by projecting the words into lower dimensions with dense space vector. A technique of CNN to extract the features of the sentence and joined these features with handcraft-features for the relation classification proposed in [24]. However, traditional CNNs are unable to hold long term semantic features. Comparatively, a unique design of CNN is the utilization of dilated convolution. It eliminates the implications of information loss, which is due to conventional down-sampling approaches in traditional pooling operations and likewise in the stride convolution. Further, it can expand the size of receptive fields at the exponential level without concerning additional parameters. Thus, it becomes feasible for the dilated convolution to capture long term semantic information.

Although there are numerous SCAs, still it appears as a collection of loosely coupled relation with social media, which is need to be strengthened. The current research is utilizing a deep learning methodology that automatically identifies information on social sensors by considering a parallel dilated convolutional neural Network (PD-CNN). From the best of our review, PD-CNN is capable of predicting appropriate information as it learns features from DS-DWRs using different distributed word representations, as mentioned above. Beyond the improvement in SCAs, several favorable hyper-parameters while utilizing DL for social media sentiment analysis are considered. Our work not only introduces the new way of social media sentiment analysis but also useful to modernize smart cities. The following are the contributions of this work:

- The generation of domain-specific distributed word representation is explored to discover the features for the enhanced exhibition of the classifier in comparison with standard methods.
- The parallelism in the dilated convolutional neural network contains three dilated convolution layers containing different dilation rates with global average pooling in a novel manner.
- Lastly, without the complex feature-engineering, the work proceeds with the employment of the DL method to provide intelligent Smart city applications through social media sentiment analysis based on the social sensor (Tweets) to augment the people's perception in smart cities.

The rest of this paper is organized as follows. Section 2 proceeded with the related work. The proposed architecture for social media sentiment analysis is presented in Section 3. While Section 4 contains the procedure of experiments. The discussion is in Section 5. The paper is concluded in Section 6.

2. Related work

Smart City and social media analytics continuous growth contributed to an advanced level in technological and urban advancements. Only a few studies in the last years clarify the complementary characteristics of the social-technical ecosystem. However, their mutual significance is not yet wholly idealized [25]. The works on sustainability which explain the smart cities by relating innovations to social networks explained in [26,27]. Social users of smart city services are well informed about the availability and productivity of these services, as described in [28]. It Implies the requirement for more practicality in smart cites research regarding social networks. For the sustainability in smart cities research, one has to concentrate on the social context related to smart cities [29]. In this Way, smart city applications (SCAs) prompt the integration of social networks and smart cities services having opinions and prospects, as explained in [30,31]. Thus, social networks have a profound impact on SCAs. Social networks contain informal methods of interactions and opinions. We attempt to analyze how these opinions contribute to promote the mentioned services. Further, these networks allowing to share status update messages. These messages have additional (meta) information, e.g., name, time, location, and hashtags. It can be an essential source that can help to provide intelligent services in smart cities. Subsequently, social network users considered as social sensors and messages as sensor information utilized in many works [32–34].

Further, "Sensing" becomes ubiquitous with the utilization of smartphones, which can effectively collect and analyze information in the form of audio, video, and text [35]. The data produced by these sensors are semantically unique when contrasted with the data generated by the physical device. This semantic data is correlative information in smart cities. Numerous works utilizing short textual messages on social networks to analyze multiple perspectives have been proposed, including event Detection [36,37] disease tracking, and monitoring [38]. One of the significances of textual data based on social networks is semantics, which is more natural for social users to understand. Among various social network platforms, Twitter data has been extensively utilized in diversified applications such as Natural Language Learning (NLP) [39], Sentiment analysis [40], and Smart cities [41].

In order to boost twitter-based sentiment classification by utilizing meta-level features, for example, strength and emotion factors indicating the polarity has been proposed in [42]. A crowd-sourcing approach of manual tagging data process on Twitter with entities such as Person, Organization, and Location proposed by [43]. This data further utilized to train the model based on Conditional Random Fields "CRF". Later on, another work of crowd-sourcing by using a similar platform introduced but fail to explain (annotate) the relations exist in the entities such as Person, Organization, and location in [44]. For tagging Twitter to recognize the emotions, a methodology classifies posts of different topics based on Joy, love, surprise, anger, sadness, and fear as positive, negative, or neutral developed in [45,46]. The rapid growth in the usage of Twitter in terms of tweets can relate to many aspects such as opinions and events. The sensitivity of these aspects restricted by the noise generated from non-informative messages, as explained in [47]. In order to get valuable information, a filter can be useful for a uniform resource locator (URL), words, mentions, punctuation, numbers, and

slang words. Also, the constraints related to a set of features must be predefined. Since the performances of such systems rely upon features quality that is collected.

To mine information from social networks, it is essential to consider machine or deep learning, and social data mining approaches, as described in [48,49]. Social media mining includes many perspectives, such as the process of collecting data related to individuals and named entities and observe their interactions to understand opinions as defined in [50]. Deep learning methods perform end-to-end representation from raw data input to useful information in many applications of smart cities. These methods work without the complicated and time-consuming hand-crafted feature engineering. In this perspective, one specific issue can resolve by employing multiple deep learning models as one DL model can be applied to deal with different problems in smart cities. These attributes influences researchers to utilize deep learning methods for analyzing the smart city data, which has sophisticated modalities such as streaming, multimedia, and especially text, which are the large amount [51–55]. In general, to extract insights from twitter based on text, the significant advantage is using unsupervised pre-training. It prevents overfitting and provides appropriate representation, especially when the dataset is not substantial.

Neural Networks such as a convolutional neural network (CNNs) and recurrent neural system (RNNs) have practiced the better outcomes for social media sentiment analysis. In this perspective, a method to improve the semantics is proposed by using RNNs variant LSTM (long-short-term memory) in [56]; however, training such networks is computationally challenging. Comparatively, CNNs found to be better in training as well as to extract high-level semantic features for the text classification based on social media [57,58]. CNNs can perform feature extraction and classification on a jointly trained task, as presented in [59]. In recent years, many researchers are sequentially using convolutional and pooling layers for the semantic modeling [60,61]. A new deep CNN architecture for short texts using two convolutional layers to deal with contextual data from character to sentence-level offered in [62]. Another variation in CNN in the form of a dilated convolution neural network (DCNN) is a better choice with the increased receptive field size. DCNN has been utilized in many works in recent years such as computer vision [63] NLP [64] and Audio processing [65].

Nowadays, DL methods convert words into the continuous vector representation as Distributed word representation. Further Learning through pre-trained representation has a significant impact on the sentiment classification based on social media data. Few works have combined pre-trained vector to improve the efficiency of semantic still the text infected with syntactic, semantic, and contextual sentiment information. To tackle these issues, there is a need to focus on the association between the words for correct prediction as proposed in [66–68]. Work on using multiple word embeddings in view of separated channels shows good results based on multi-sentence classification introduced in [69]. However, multiple embeddings dimensions have to be the same, which limits the proposed model scope as pre-trained representations can exist widely in different dimensions.

For SCAs, A CNN based model employing multiple factors, for example, position, and entity to recognize the event from sentence related to social media data explained in [70]. Another work on sentence-level classification using social media data suggests that a sentence may have two or more events/opinions proposed by [71]. Based on this, they introduced a dynamic multi-pooling layer specified for the particular opinions related to some events to extract more valuable information.

Although many researchers use CNN, still CNNs is not capable of handling long term semantic features over input sequence. Further, to capture long term semantic feature, CNN needs to rely on stacking multiple layers for convolution and pooling. To address the issues, a dilated convolution neural network (D-CNN) followed by domain-specific distributed word representation is utilized. From the best of our knowledge, this fusion helps the underlying stage, which is fundamental for SCAs.

3. Proposed architecture for social media sentiment analysis

This section presents a theoretical model of the proposed method for social media sentiment analysis based on twitter for SCAs. It has domain-specific distributed word representation (DS-DWR) comprised of different distributed word representations. The D-CNN architecture consists of three parallel dilated convolution neural network layers with global average pooling (GAP) to capture long term semantic features. We employ two-dimensional convolution operations to attain improved performance. In this way, the words of the datasets are initialized to the PD-CNN as a matrix having two dimensions. The dilated convolution operations on this approach help to extract additional essential features as well as support the proposed model toward the designed DS-DWR. Our proposed design has succeeding portions: Domain-specific distributed word representation, three parallel dilated convolution layer, Global average pooling, and output layer, as shown in Fig. 1.

3.1. Domain-Specific Distributed Word Representation (DS-DWR)

Distributed Word Representation is an essential element to represent the words as dense vectors. Feature extraction using these representations is valuable that leads to improve the execution of social media sentiment analysis. Many techniques utilized multiple representations in text classification described in [72–74]. However, no technique solely adequate to address semantic and contextual sentiment information of social media data. In this section, a correlation between different learning such as Word2vec [21], fastText [22], GloVe [23], and together with their concatenation is handled as pre-processing.

These different distributed word representations could be directly concatenated to form enriched representations. Such concatenation not just influences various techniques in term of increasing lexical data but additionally deal with the out of vocabulary (OOV) words. A weighted mechanism is used for different representations. It is observed that different weights help to analyze different representations, so higher weights are considered for feature selection. The process of concatenation with associated weights theoretically helps in sentence encoding. We suggest training of concatenated distributed word representations with 100 and 200 dimensions. We turn characters into lowercase and discard words that exist not up to ten times while the optimum size of context window 5 is used. Further, in the course of training of our DS-DWR, the learning rate is dropped symmetrically as with progress in training, the early appeared text seems to have a more significant effect on the model. We train the embeddings on two datasets separately, the first dataset based on twitter related to real-world events employed by [75].The other dataset accumulated through TAGS by following the work of [76].

3.2. Dilated Convolutional Neural Network (D-CNN):

In social media sentiment analysis, the classical methods dependent on CNN are restricted towards the size of convolutional kernels, which causes two primary issues. First, CNN is merely capable of capturing short term dependencies in terms of semantics. The second one is significant increase of parameters as it is relative to the extension of convolutional kernels. To handle these issues, a dilated convolutional Neural Network (D-CNN) is introduced in [63]. It is the improved version of CNN, which introduces another hyper-parameter to the receptive layer. D-CNN raises the overall network's performance by increasing the size of the receptive field to capture additional information by placing zeros in the filter components. The increased size of the receptive field is considered essential for the prediction and classification tasks. A one-dimensional D-CNN with dilation "l" which convolves signal "F" along with kernel "k" consisting of receptive field size "r" indicated in [77]. This can moderately extend to the two-dimensional dilated convolution as in the following expression:

$$\left(F *_{d_i} k\right)_t = \sum_{\tau} k_\tau F_{t-d_i\tau} \tag{1}$$

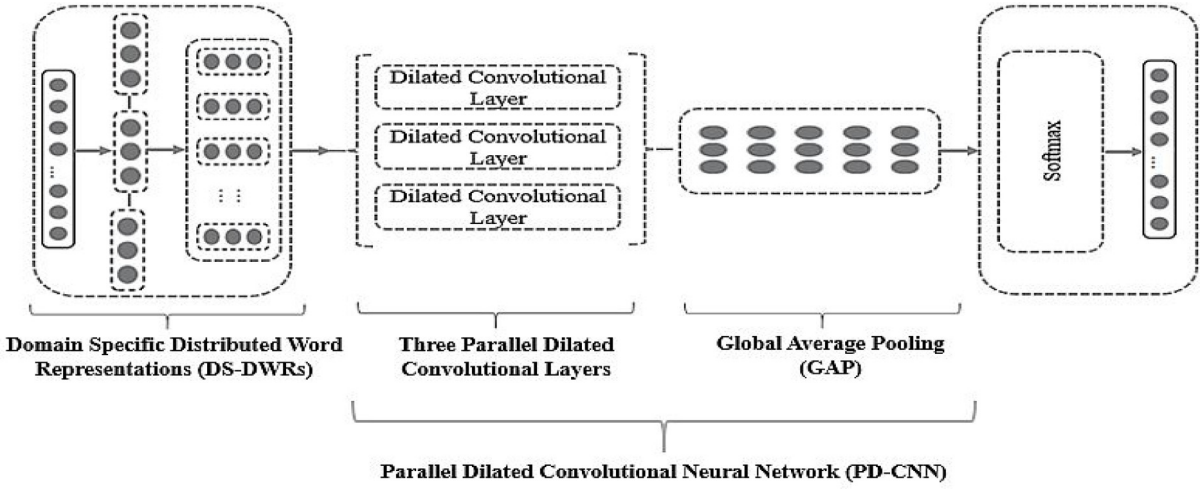


Fig. 1. Proposed PD-CNN with GAP followed by domain-specific distributed word representation.

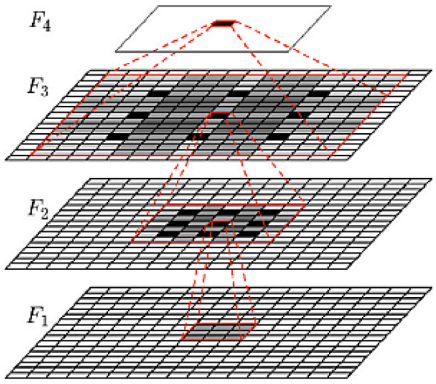


Fig. 2. Example of three dilated convolution layers [77].

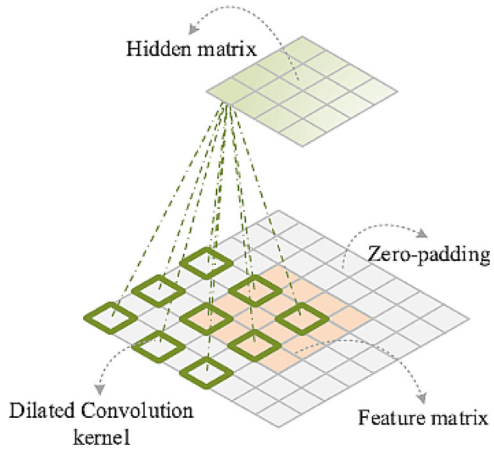


Fig. 3. Dilated Convolution with dilated rates $(2 * 2)$ [78].

where $*_{d_i}$ represents d_i dilated convolution. A model is comprising of three dilated convolutional layers can be found in Fig. 2, where the factor of dilation grows up significantly at each layer. The Fig. 2 shows an example of three dilated convolution layers that have a particular feature map represented as (F_1, F_2, F_3, F_4) utilizing a respective dilation convolution as $(1, 2, 3, 4)$ and size of receptive field related to each element as $((3 \times 3), (2^3 - 1) \times (2^3 - 1), (2^4 - 1) \times (2^4 - 1))$.It can be seen the size of the receptive field related to each element in F_{i+1} as $(2^{i+2} - 1) \times (2^{i+2} - 1)$.

Parallel Dilated convolution Layers: In these layers, dilated convolutions are implemented to extract the semantics with the exponential growth in the size of the receptive field without compromising on dimensions. Further, these empowers the association of distant elements without expanding the parameters. The dilated operation by employing different dilation rates increase the convolution kernel over several scales. In this work, parallel dilated convolutions of different dilation rates over different scales is employed in which rates regarded as hyperparameters. A simple design of dilated convolution with dilated rates [78] is shown in Fig. 3. Furthermore, there are probabilities of inappropriate information for capturing semantics at long distances. Therefore, we selected [1,2,4] as dilation rates, so the highest layer not merely process information at the long distance and diminishes the impact of containing insignificant data. Our model is capable of extracting the long term semantic features by utilizing low dilation rates in the interest of words and a sentence through significant dilation rates. The computation of three parallel dilated convolution layers for the extraction of high-level features exhibited in the accompanying expressions that are considered as the input to the following layer:

$$D_c^1 = RRELU\left(F *_{d_1} X_t^x\right) \tag{2}$$

$$D_c^i = RRELU\left(F *_{d_i} D_c^{i-1}\right) \tag{3}$$

$$D_c^l = RRELU\left(F *_{d_l} D_c^{l-1}\right) \tag{4}$$

where $F *_{d_i}$ is d_i Dilated convolution in the respective layer "l" of exponential increase dilation width.

3.3. Global average pooling layer

A global average pooling (GAP) operation is applied to the output produced by three parallel dilated convolutional layers. GAP additionally replaces the fully connected layer because it coordinates in feature maps and a classification layer. Besides, It would not require any configuration of parameters. GAP focused on the average value associated with each feature map instead of involving a fully connected layer related to a particular feature map. This average value consists of a feature map activates a specific value for each class, and subsequently, the vector passed into the softmax function. The following expression shows the probability distribution over social media sentiment analysis.

$$P\left(i_j | k, \theta\right) = \frac{\exp\left(x_j\left(k, \theta\right)\right)}{\sum_{1 \le i \le |X|} \exp\left(x_i\left(k, \theta\right)\right)} 1 \le j \le |X| \tag{5}$$

Table 1
Selection of distinct values related to hyper-parameters.

Activation function	No. of hidden layers	No. of neurons in hidden layer	Optimization algorithm	Mini batch size	No. of epochs	Dilation filter size	Dilation width
RELU	5	90	SGD	32	10	2, 2, 2	1, 2, 4
Tanh	10	185	ADAM	64	50	2, 2, 3	
R-RELU	15	250	RMSprop	128		2, 2, 4	

Where, $x_j(k,\theta)$ as average pooling result with a parameter set θ corresponding to the class j, and class space is represented as X. For the minimization of negative log-probability, stochastic gradient descent is exploited. Softmax loss is a loss function which explains a mixture consisting of multinomial logistic loss and softmax. We utilized softmax loss as a base loss function in our work. The concept of softmax loss is to make sure the model authenticity towards authentic classification. The softmax typically transforms the predictions in positive values as well as distribute them for a probability function. Such probabilistic distributions are utilized to calculate the overall softmax loss, as follow:

$$L_{soft\max} = -\frac{1}{N}\left[\sum_{w=1}^{N}\sum_{x=1}^{K} 1\left\{y^{(w)} = x\right\}\log i_j^{(w)}\right] \tag{6}$$

4. Experiments

In this section, the datasets and experiment setups are presented firstly, and then experimental results and analysis are depicted. Finally, distinct approaches are compared to selected datasets. The objective to examine the considered techniques; our work undergoes an effective classifier based on D-CNN followed by DS-DWR.

4.1. Datasets

To precisely assess techniques on multiple corpora with various domains and size, we test the system on multiple datasets for its applicability, efficiency, and reliability. We used two datasets separately, the first dataset based on twitter that contains a range of different real-life events, like stories, emergencies, news and sports. This Dataset we tend to use is available publically [79]. The total number of tweets that were collected is 810 from different 27 events, each of 30 tweets. The other dataset accumulated through TAGS [80]. It is a twitter search interface (API) that works on search terms given by the user through query and stores the results on a free provided google sheet template. Through TAGS, the user can manually put the query. It also provides setup tags to update the sheet at any time when required. Tweets were collected using "smart cities and social media" as search terms for one month accumulate to 7481. The few pre-processing performed on the datasets. The utilization of core NLP offered by Stanford for tokenization and lemmatization, as described in [81]. The short sentences up to four words, as well as stop words, are removed. To execute better and for reliable analysis, we decide to shuffle the texts in both datasets as it effect of all text to be equivalent [82]. We acquired three sets for both datasets, where each dataset contains an equal number of Randomly Shuffled Tweets named RST-1, RST-2, RST-3.

4.2. Experimental setup

We augmented our experimentations on chosen datasets by utilizing multiple distributed word representations and concatenate them for domain-specific distributed word representation. Our experiments examine different possibilities and appropriate parameters. These include activation functions, number of neurons, hidden layer, epochs, size of size, optimization algorithm, and filter size for the proposed methods shown in Table 1. The activation function typically sets the output, which acts as input for the neurons in the succeeding layer. The rectified linear unit (RELU), and hyperbolic tangent (tanh) are the most considered activation functions [83]. They handle the problem of vanishing gradient as it bound the gradients for being too small during training. Further, we employed Randomized-RELU "R-RELU" wherein the parameters concerning adverse effects are randomly sampled as regular distribution in training and reduces overfitting [84]. The width of the model relies on the number of hidden layers that influence the overall computational complexity of the network. Accordingly, we investigated our architecture by considering 5, 10, 15 in terms of hidden layers.

On the other hand, the number of hidden layers has no concept for a definite number of neurons in hidden layers described in [85]. The inadequate selection of the number of neurons in hidden layers leads to problems since few neurons cause underfitting, while several neurons cause overfitting that eventually affects model training. Keeping in mind, we use 90, 185, 250 neurons in hidden layers. For training, stochastic gradient descent SGD is employed, having a learning rate of 0.01. Additionally, a methodology for stochastic optimization "ADAM" also included by using a learning rate of 0.001 which needs simple tuning compared to SGD. The training performance can be enhanced by considering the root mean squared propagation "RMSprop" optimizer. RMSprop accumulates gradients in a fixed-window as opposed to collecting all the gradients for a momentum.

Moreover, the training of the neural networks is feasible via mini-batches, which splits the massive datasets to smaller parts. Therefore, mini-batch gradient descent is considered with batch sizes of 32, 64, and 128 correspondingly. In order to hold the average output consisting of vectors regarding each word "time step", a global average pooling "GAP" is attached, over each batch. Furthermore, the output associated with preceding processes is also concatenated. The number of epochs refers to the number of times the full dataset passed through the neural network for generalization. Like other hyper-parameters, numbers of epochs "Few or More" cause underfitting and overfitting, respectively. For the best assessment by using the above hyper-parameters, we choose the number of epochs from 10 to 50. A D-CNN with a different filter size consists of (2, 2, 2/2, 2, 3 /2,2,4), and the width of dilation is 1, 2, 4 used to process the high-level features obtained from 3 parallel dilated convolution PD-CNN.

4.3. Experimental results and analysis

Our experiments are performed on various evaluation metrics that are achieved on our datasets by referential and proposed methods. Since accuracy $\frac{TP+TN}{P+N}$ is classical metric in many deep learning techniques. However, this metric found to be inappropriate in the case of imbalanced data. Therefore, in this case, precision $\left(\frac{TP}{TP+FP}\right)$ and recall $\left(\frac{TP}{P}\right)$ are more reliable metrics. Further, F1-Score is employed to examine the equilibrium in recall and precision. A processor of 8 cores with 2000 MHz and a RAM of 16 GB is used to perform all the experiments. An open-source ML library for research and production TensorFlow [86] is engaged for the training and evaluating the proposed architectures.

We endorsed the evolution of our work in consideration of referential methods, which encompassed every model with pre-trained word vectors and compared to proposed architecture. For the Comparison, few assessments with individual pre-trained distributed word representations on the same PD-CNN with standard hyperparameters are performed. We evaluate the baseline model on accuracy and F1-Score. The most elevated accuracy achieved is 72.34% with F1-Score 67.24% by using FastText as compared to the accuracy of 71.45% with F1-score

Table 2
Accuracy achieved by baseline methods of all datasets.

Methods	RST-1	RST-2	RST-3
GloVe-PD-CNN	71.45%	70.88%	67.98%
Word2Vec-PD-CNN	69.52%	71.32%	68.93%
FastText-PD-CNN	72.34%	69.52%	70.76%

Table 3
F1-Score by baseline methods on all datasets.

Methods	RST-1	RST-2	RST-3
GloVe-PD-CNN	65.46%	64.58%	58.15%
Word2Vec-PD-CNN	60.52%	65.46%	58.23%
FastText-PD-CNN	67.24%	60.52%	64.62%

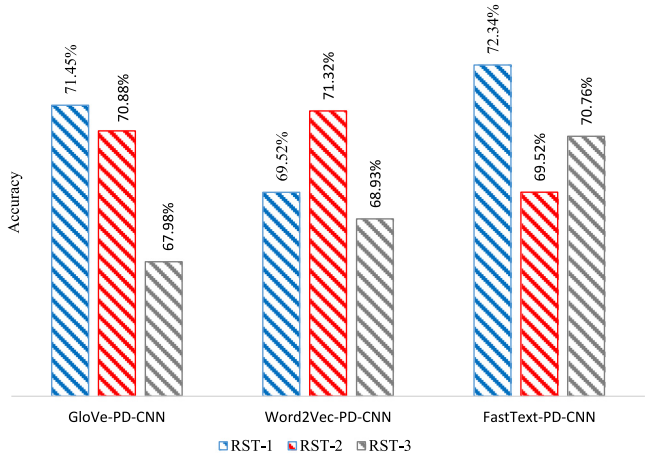


Fig. 4. Baseline methods accuracy on RST-1, 2, 3.

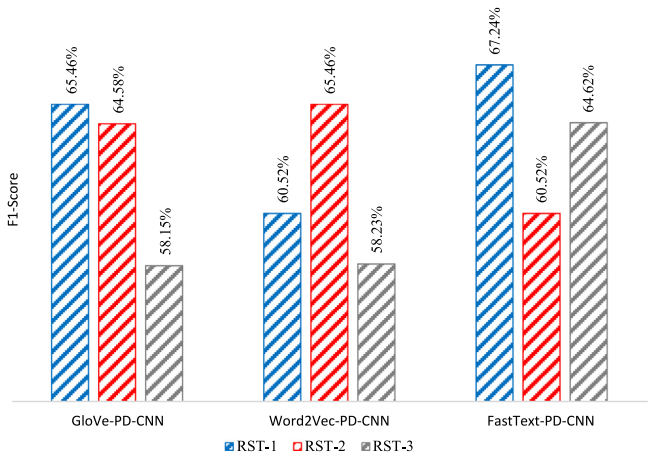


Fig. 5. F1-Score on baseline methods using RST-1, 2, 3.

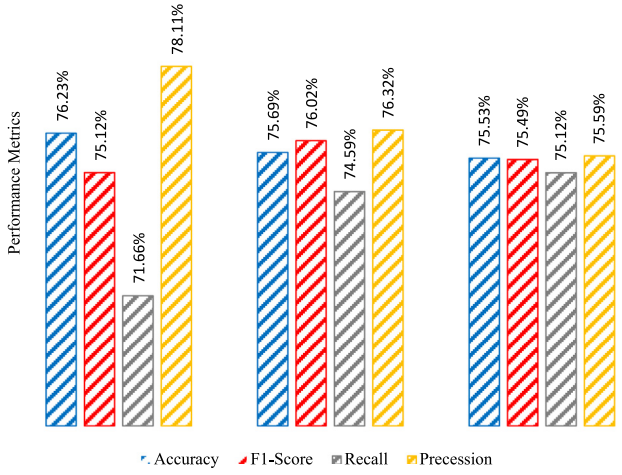


Fig. 6. Proposed Method Accuracy, F1-SCORE, Recall, and Precision on RST-1.

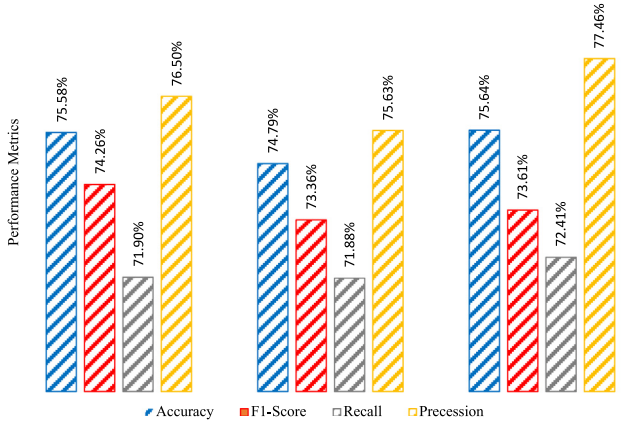


Fig. 7. Proposed Method Accuracy, F1-SCORE, Recall, and Precision on RST-2.

65.46% by using GloVe On RST-1 are presented in Tables 2 and 3. The referential methods evaluation metrics are shown in Figs. 4 and 5.

The accuracies of the proposed methods mentioned in Tables 4–6 with selected hypermeters and their alternatives for the best evaluation, as mentioned in Table 1. The utilization of more hidden layers does not result in efficient improvement. However, surprisingly, the optimization through RMSprop produces appropriate precision 78.11% with Randomized Rectified Linear Unit (R-RELU). The other architectures optimized through SGD and ADAM accomplished slightly lower precision 76.50%, 77.90% with RELU and Tanh, respectively. The evaluation metrics consists of F1-Score, recall, and precision of the proposed model can be seen in Tables 7–9.

Additionally, the batch sizes in our architecture, denying many works that are related to the best performance with the batch sizes consisting of 2 and 32. Similarly, the best accuracy obtained on our datasets is 76.29% and 75.88%, with batch sizes of 64 and 128, respectively. Furthermore, 15 hidden layers are considered for the best architecture. On account of the best recall of our architecture is 75.64% on RST-3 in Table 9. The Overall results in terms of accuracy, F1-score, recall, and precision are shown in Figs. 6–8. These results show that our proposed architecture has excellent flexibility in exploiting different metrics.

5. Discussion

Currently, the improvement of deep learning approaches promoted the accessibility of distributed word representations. In this work, our investigation related to the quality of the representations to execute social media sentiment analysis for smart cities. We observed various evaluation metrics and hyperparameters, as mentioned in Table 1. One of the main challenges is consists of various dimensions of distributed word representation as our implementation has different dimensions. Different distributed word representations found unique to create Domain Specific Distributed Word Representation (DS-DWR) through a weighted mechanism. The development of a new structure based on a parallel dilated convolution neural (PD-CNN) network is utilized to use large receptive fields. It includes three parallel dilated convolutional layers rather than stacking of multiple dilated convolutions for sequential processing of textual data. PD-CNN just like WaveNet, does have

Table 4
Accuracy achieved by proposed method on RST-1 with distinctive hyper-parameters.

Datasets	Accuracy	Batch size	No of epochs	No of hidden layers	No of neurons	Optimization algorithm	Activation function
RST-1	76.23%	128	10–50	15	250	RMSprop	R-RELU
RST-1	75.69%	32	10–50	5	90	SGD	RELU
RST-1	75.53%	64	10–50	10	185	ADAM	Tanh

Table 5
Accuracy achieved by proposed method on RST-2 with distinctive hyper-parameters.

Datasets	Accuracy	Batch size	No of epochs	No of hidden layers	No of neurons	Optimization algorithm	Activation function
RST-2	75.58%	32	10–50	5	90	SGD	RELU
RST-2	74.79%	128	10–50	15	250	RMSprop	R-RELU
RST-2	75.64%	64	10–50	10	185	ADAM	Tanh

Table 6
Accuracy achieved by proposed method on RST-3 with distinctive hyper-parameters.

Datasets	Accuracy	Batch size	No of epochs	No of hidden layers	No of neurons	Optimization algorithm	Activation function
RST-3	75.88%	64	10–50	10	185	ADAM	Tanh
RST-3	76.29%	128	10–50	15	250	RMSprop	R-RELU
RST-3	75.48%	32	10–50	5	90	SGD	RELU

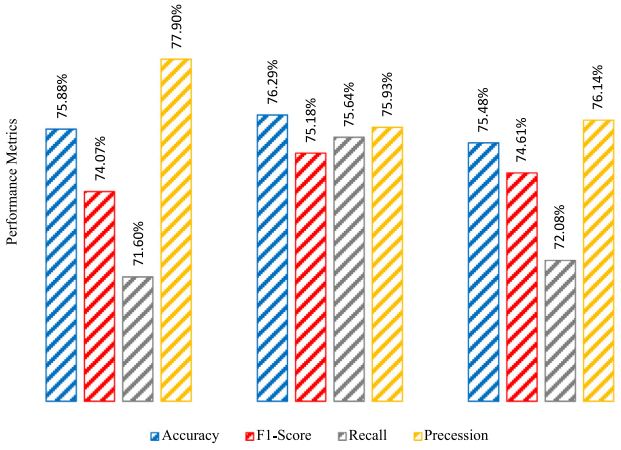


Fig. 8. Proposed Method Accuracy, F1-SCORE, Recall and Precision on RST-3.

Table 7
Performance metrics of proposed method on RST-1.

Datasets	F1-Score	Recall	Precision
RST-1	75.12%	71.66%	78.11%
RST-1	76.02%	74.59%	76.32%
RST-1	75.49%	75.12%	75.59%

Table 8
Performance metrics of proposed method on RST-2.

Datasets	F1-Score	Recall	Precision
RST-2	74.26%	71.90%	76.50%
RST-2	73.36%	71.88%	75.63%
RST-2	73.61%	72.41%	77.46%

Table 9
Performance metrics of proposed method on RST-3.

Datasets	F1-Score	Recall	Precision
RST-3	74.07%	71.60%	77.90%
RST-3	75.18%	75.64%	75.93%
RST-3	74.61%	72.08%	76.14%

a large size of the receptive field through convolutions with dilations to achieve long-term dependencies [65]. Also, have the advantage of computational time which is linearly determined by the length related to sequences. Further, we engage Global Average pooling (GAP) as opposed to mean pooling considering the way that particular pooling is better in the circumstance of position-invariance. A combination of D-CNN with specific pooling seemed to improve the performance in comparison with [29], which utilized dilated convolution without pooling operation. By evaluating the general and domain-specific distributed word representation (DS-DWR), we observe that setting up DS-DWR is significant despite having the limited size of the corpus. Our experiments assist essential revelations for the implementation of PD-CNN in terms of (i) different distributed word representations to generate DS-DWR with GAP to classify text using the semantics of social media for smart cities application (SCAs). (ii) Hyper-parameters preference and configurations that have a primary impression on overall realization.

Nowadays, social networks such as Twitter should be analyzed as it provides micro-content data that have phenomenal impacts on decision marking. Twitter is probably the commonly used social network by which people can share information regarding events and their opinion. Several techniques in current literature have been examined; perhaps the problems still exist, i.e. semantic refining, availability of sentiment analysis to mine opinions, the procedures to analyze social media data, and the utilization of deep learning approaches in the interest of SCAs. In this consideration, social media mining seems to have an excellent capability to provide the usage of social media data analysis for reliable services based on opinions that are shared through the social network. Our approach in this paper is associated with a social media mining algorithm that is capable of mining opinions. An exciting development is to concentrate on textual data that describe different opinions related to any instance referred to as social media sentiment analysis. It is an increasing solution to perceive the feelings of human actors on social networks. Several SCAs have merits of sentiment analysis of social media data such as twitter. These applications utilize the opinions of users regarding any instance, such as happiness, security, and events.

One of the motives of this paper is associated with the authorities that are necessary to realize the continuing process related to the evolution of technology. The smart city techniques possess a direct relation with the availability of reliable, robust, and value-adding technologies. Thus, our proposed study promotes smart city authorities to activate new techniques for social media data mining. A more in-depth direction for smart city professionals to recognize the scalability of techniques together with implications. It is time to standardize SCAs, which will promote the significance of social media data mining. However, the intuitions to utilize social media data mining in smart

cities incorporate a continuous increase in the size of social media data. Thus, our model can be considered as an essential resource for SCAs to multiple disciplines. We can articulate that without examination of social media data and a neural network-driven approach, SCAs will perform unsatisfactorily. By using social media data, together with neural networks, a more respondent, economical, as well as proactive SCAs, can be achieved.

6. Conclusion

Social media has developed into an essential mean for communicating the information. This paper interprets various hyper-parameters configurations for neural networks to predict and classify sentiment words on social media for SCAs. It enables smart city authorities to optimize the model to maximize distinctive evaluation metrics. We first employ domain-specific distributed word representation (DS-DWR) trained on our specific social media data that outperform generic distributed representation trained on a particular corpus. Further, we suggest three parallel dilated convolution neural network (PD-CNN) layers with global average pooling (GAP). PD-CNN is augmented with DS-DWR performs well on tweets for picking informative messages over long term semantic sequences. The experimental results confirm that our model can perform better compared to baseline models. However, it has observed that extensive training data increase overall performance — also, the imbalanced aspect of training data in our work remains a challenge. Further, our approach is restricted to a single-subject based collection of tweets through relevant keywords that can be expanded in future work. Our work motivates smart city explorers to consider the current research and proposed more inventive and scientifically useful deep learning-based models for SCAs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

L.V. Yunrong: Supervision.

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